De la cimentoplastie à la cryoablation et à l'ostéosynthèse percutanée : état de l'art en radiologie interventionnelle osseuse

Titre original : Cementoplasty to cryoablation: review and current status

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Synthétise la stratégie multimodale appliquée à Cochin : associer destruction tumorale (cryoablation/RF) et consolidation mécanique (cimentoplastie / vissage) en une seule séance.

2. Synthèse Clinique en Français

Résumé exécutif : Revue générale (*BJR Open*, 2024) détaillant l'évolution des gestes percutanés osseux : critères de sélection entre cimentoplastie seule, ostéosynthèse par vis canulées et association ablation thermique + consolidation selon la classification de Mirels et le score SINS.

Analyse détaillée : Cet article propose un algorithme décisionnel complet pour le radiologue interventionnel ostéo-articulaire. Il précise les propriétés biomécaniques du ciment PMMA (excellente résistance aux forces de compression au rachis et au toit du cotyle, mais résistance limitée aux forces de torsion et de traction sur le bassin postérieur et les os longs, justifiant l'association au vissage percutané). Il détaille également la chronologie optimale lors des gestes combinés : réaliser la cryoablation suivie d'un réchauffement complet de la loge avant d'injecter le ciment PMMA afin de ne pas altérer sa polymérisation.

3. Protocole Technique & Paramètres Opératoires

Évaluation biomécanique pré-opératoire (score SINS au rachis, localisation en zone portante au bassin) • En cas de geste combiné Cryo + Cimentoplastie : attendre la décongélation complète du glaçon avant d'injecter le PMMA.

4. Points Clés à Retenir pour la Pratique

- Le ciment PMMA résiste parfaitement à la compression (corps vertébral, toit du cotyle) mais nécessite des vis canulées en zone de cisaillement (anneau pelvien, col fémoral).
- Toujours attendre la fonte du glaçon après cryoablation avant d'injecter le ciment pour éviter une polymérisation retardée ou des fuites.
- La cryoablation préalable réduit la pression intra-tumorale et diminue le risque de fuite veineuse ou épidurale de ciment.
- L'évaluation multidisciplinaire (radiologue interventionnel, orthopédiste, oncologue, anesthésiste) optimise le calendrier vis-à-vis de la radiothérapie.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Recent advances in percutaneous image-guided techniques have empowered interventional radiologists with diverse treatment options for the management of musculoskeletal lesions. Of note, there is growing utility for cementoplasty procedures, with indications ranging from stabilization of bone metastases to treatment of painful vertebral compression fractures. Likewise, cryoablation has emerged as a viable adjunct in the treatment of both primary and secondary bone and soft tissue neoplasms. These treatment options have been progressively incorporated into the multidisciplinary approach to holistic care of patients, alongside conventional radiotherapy, systemic therapy, surgery, and analgesia. This review article serves to outline the indications, technical considerations, latest developments, and evidence for the burgeoning role of cementoplasty and cryoablation in the musculoskeletal system, with an emphasis on pain palliation and tumour

control.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Introduction

Over the last few decades, various percutaneous image-guided ablation and stabilization techniques, including cryoablation and cementoplasty, have emerged in the management of musculoskeletal lesions. There is growing utility of these procedures especially in the context of malignancy, where bone pain is one of the most common types of pain, and the skeleton is the third most frequent site of metastases.1 This is compounded by the prevalence of bone metastases, with the annual age-adjusted incidence rate of newly diagnosed bone metastasis approximately 18.8 per 100 000 in the United States.2

Currently, radiotherapy is considered the preferred treatment option for palliation for uncomplicated metastatic bone pain. However, radiotherapy alone does not contribute to bone stability.3,4 Other limitations of radiotherapy include radio-resistance of some tumours, risk of radiation-related fracture especially to weight-bearing bones, osteitis, osteonecrosis, as well as radiation dose limits for 1 body site.5,6 Additionally, radiotherapy has a delayed pain relief onset, typically 4 weeks and occasionally up to 15 weeks.7

Surgery is only indicated as first-line treatment for selected metastatic bone lesions, such as slow-growing tumours and patients with relatively good prognosis.8 Surgical decompression and instrumentation have a primary role in the management of acute spinal instability and spinal cord compression with neurological deficits.4 While opioids are recommended for management of cancer-related bone pain, they are associated with multiple common side effects and sometimes suboptimal pain relief.9,10

Given the limitations of traditional strategies, this review article serves to provide an update on the increasing role of cementoplasty and cryoablation in the musculoskeletal system, with an emphasis on pain palliation and tumour control.

Cementoplasty

Percutaneous cementoplasty is a minimally invasive technique comprising of cannula placement within a bone lesion with subsequent cement injection.11 The cement, usually polymethylmethacrylate, hardens in consistency after about 10-20 min. Cement polymerization and hardening involve an exothermic reaction, with temperature peaks of up to 75°C, and play an accessory analgesic role through the destruction of adjacent nociceptors.12

Indication

When considering cementoplasty for bone metastases, a multidisciplinary team including interventional radiologists, radiation oncologists, medical oncologists, and orthopaedic surgeons should be engaged. Cementoplasty is typically considered in the following clinical scenarios: persistent pain or imaging evidence of tumour progression despite maximized radiation therapy, contraindications to radiation therapy, lack of patient desire for radiation therapy, and/or inadequate treatment response to systemic therapies and analgesia,13 as well as prophylactic stabilization for impending pathological fractures in the axial skeleton.14

Preprocedural cross-sectional imaging is essential to determine the number and site of lesions, needle size, and approach, as well as measurements for ablation zone (if performed). MRI imaging features may also be useful to guide patient and vertebral level selection; levels with marrow oedema correlating with pain should be targeted for optimal pain relief.15 Careful review of prior imaging, with a low threshold for repeating the patient’s cross-sectional imaging, is advised, especially if new or significantly worsening symptoms are reported since the latest imaging was performed.16 In the authors’ experience, preprocedural imaging is usually current, preferably within a few days of procedure, and at most within a month from procedure.

Severity of the spinal instability is an important consideration and can be evaluated via various scoring systems, including the Spinal Instability Neoplastic Score (SINS).17 Surgical consultation for potential tumour

resection or debulking with stabilization is suggested for patients with a SINS of 7 or higher. Spinal metastases resulting in central canal stenosis are typically managed with surgical intervention.18

As regards the spine, suitable patients for vertebroplasty include those with chronic pathological fractures refractory to radiotherapy or other conservative measures, acute vertebral fractures with less than 6 weeks of symptoms, and painful pathological vertebral fractures without spinal cord compression or significant neurological deficit.19 The most appropriate timeframe for substantial pain relief may be 2-4 weeks from onset of injury, as suggested by a meta-analysis.20 However, treatment in the initial 4 weeks of vertebral fractures may lead to a higher risk of complications, including cement leakage.21,22 In the authors' experience, vertebroplasty can be performed before or after radiotherapy and readily coordinated during multidisciplinary discussions.

Cementoplasty is contraindicated in patients with coagulopathy, unstable spinal lesions, presence of local or systemic infections, allergy to bone cement, and asymptomatic vertebral compression.23,24 Vertebral plana with 90% or greater height loss and extensive osteolytic destruction especially at the posterior cortex of the target vertebral body are relative contraindications.25

Technique

In the authors' experience, cementoplasty is usually tolerated well under conscious sedation with local anaesthesia; general anaesthesia can also be considered for certain patients. Patients are positioned according to the site of target lesion; in conventional vertebroplasty, this usually means a prone position with padding under the thorax and ankles. Local anaesthesia is administered along the puncture site and expected cannula trajectory, infiltrating the subcutaneous tissues, muscles, and pain-sensitive periosteum.

For vertebroplasty of a lumbar segment, a transpedicular approach is preferred where the upper outer aspect of the pedicle is accessed. The cannula is directed medially through the pedicle and into the anterior third of the vertebral body to the midline inferiorly. This optimizes cement distribution to ideally include most of the vertebral body with cement crossing the midline (Figure 1).19 A costotransverse approach for the thoracic vertebra is also feasible due to smaller pedicle sizes. Bipedicular access may be considered in some scenarios including suboptimal positioning of a single cannula, or severe central vertebral height loss preventing cannula placement along the midline. Other novel access routes, including transoral access of C2 and a trans-discal approach, have been reported.26,27

Two-level vertebroplasty for osteoporotic lumbar compression fractures. Intra-procedural fluoroscopic images obtained in frontal (A) and lateral (B) planes to visualize cement deposition in the vertebral bodies. Postprocedural cone-beam CT images in coronal (C) and sagittal planes (D) demonstrated satisfactory cement fill with no leakage.

Aside from vertebroplasty, other percutaneous vertebral augmentation procedures include balloon kyphoplasty and vertebral stenting. During kyphoplasty, a balloon-like device is inflated inside the vertebral body, and its expansion restores vertebral body height, creating a cavity into which cement is then injected.24 Patient positioning and vertebral body access are similar to conventional vertebroplasty. A reamer is inserted over the access cannula, through which the stenting device or inflatable balloon is inserted. Once deployed, an intravertebral cavity is created with partial restoration of vertebral height.

Cementoplasty has also been performed at extraspinal positions, notably the sacrum, acetabulum, and long bones. For percutaneous sacroplasty (Figure 2), both a short-axis or long-axis cannula placement technique can be adopted depending on the size and extent of the fracture or lesions. The short-axis technique is better suited under CT guidance, with the cannula advanced perpendicular to the dorsal surface of the sacral ala. However, the volume of injected cement at each cannula may be limited, necessitating the placement of additional cannulas or adopting a more oblique approach.28

Percutaneous sacroplasty. (A) MRI axial T2W with fat suppression image revealed marrow oedema in the sacral ala bilaterally (arrows) due to insufficiency fractures. (B) An oblique approach of cannula placement was planned on CT (arrow). Under CT guidance, bone cannulas were advanced into the sacral ala bilaterally (C and

D), lateral to the sacral foramina. (E) Fluoroscopic control image in the sagittal plane was obtained prior to cement injection. Postprocedural images in the lateral (F) and frontal (G) projections demonstrated satisfactory cement fill with no cement leakage.

For percutaneous acetabuloplasty (Figure 3), either a posterior or anterior approach can be adopted. Cement is deposited along the acetabular roof and posterior column, as they form part of the load-transmission axis. Displaced fractures with acetabular protrusion are contraindications due to the risk of intra-articular cement leakage.

Acetabular cementoplasty for a patient with metastatic lung cancer with persistent left hip pain. (A) Preprocedural CT coronal image revealed lytic lesions involving the left acetabular roof (arrows), while axial CT image (B) revealed sclerotic components in the iliac bone (arrowheads), suspicious for bony metastases. Under CT guidance with needle guidance software (XperGuide, Philips, Netherlands), acetabular cementoplasty was performed with 16 mL of cement deposited along the acetabular roof, preferentially at the posterior column as confirmed on axial (C), sagittal (D), and coronal (E) reformats.

Imaging guidance

In the authors' experience, cannula placement is usually performed under fluoroscopy, cone-beam CT, or conventional CT, sometimes with the assistance of advanced navigation software and real-time multiplanar reformats, depending on resource availability and operator preference. CT is preferred for cervical and upper thoracic spinal access due to the excellent spatial resolution of adjacent soft tissue structures. CT gantry angulation or needle guidance software are useful adjuncts in difficult-to-access lesions. Ultrasound is useful in certain situations such as visualizing critical neurovascular structures close to the cannula trajectory. Fluoroscopic guidance for cement injection is preferred to visualize real-time cement flow and distribution.

As musculoskeletal interventional oncology procedures become increasingly complex, many procedures are performed under a combination of CT and fluoroscopic guidance. This allows the acquisition of excellent spatial resolution on the axial plane with CT and real-time visualization of cement injection or embolization under fluoroscopy. The newest cone-beam CT units have incorporated artificial intelligence capabilities including deep learning algorithms that allow guidance of bone access cannulas without requiring fluoroscopy, markedly reducing radiation dose.31 MRI guidance has also emerged, allowing target identification in soft tissue lesions while avoiding ionizing radiation,32,33 although its usage is limited by costs and availability.

Effectiveness

Previous trials such as VERTOS I and II trials demonstrated the benefit of vertebroplasty over conservative management for osteoporotic vertebral fractures.34,35 However, 2 large prospective multicentre studies and a subsequent Cochrane review failed to show a difference in pain improvement compared to a sham procedure, which involved injection of local anaesthetics but without cement deposition.36–39 The VERTOS IV trial replicated this conclusion in acute fractures up to 9 weeks.39 Major criticisms of these trials include a heterogeneous and less than ideal patient selection, heterogeneous and methodological flaws in the calculation of response to pain, underpowered study, and varying and suboptimal cement volume injected. In addition, the sham procedure of periosteal local anaesthesia at the dorsal pedicle was considered active treatment with significant effects on medial branch and sinuvertebral nerve blockade.40 Subsequently, the VAPOUR trial showed benefit over placebos in patients with acute fractures and symptom duration of less than 6 weeks.41

A follow-up of patients recruited for the VERTOS II and IV trials revealed statistically significantly more patients had a high pain score at 12-month follow-up in the sham and conservative group, as compared with the vertebroplasty group.42 Patients with moderate fracture deformity were also less likely to have high pain scores if they were treated with vertebroplasties.42

Pathological vertebral compression fractures are the second most common indication for vertebroplasty. A retrospective observation study of patients with metastatic compression fractures who underwent vertebroplasty reported improvement of visual analog scale (VAS) pain score from 5.8 to 2.7; patients with posterior column involvement and paravertebral extension reported greater pain relief.43 The efficacy of

vertebroplasty on pain reduction and disability improvement is also well documented in multiple myeloma patients, with sustained results on long-term follow-up (5 years).44 Pain relief can be substantial; a study documented median VAS scores decreasing from 9 to 1 postvertebroplasty.45 Vertebroplasty is also a potentially effective percutaneous technique to treat symptomatic intraosseous haemangiomas, with lasting results.46

Currently, the overall consensus is that vertebral augmentation is appropriate for painful vertebral compression fractures refractory to conservative therapy, as well as for select neoplastic vertebral lesions.47 The authors are concordant with these recommendations. In the authors' experience, vertebral augmentation for symptomatic neoplastic vertebral lesions with multidisciplinary input tends to yield favourable results. In addition, there are potentially additional benefits of vertebroplasty aside from pain relief and mortality which require further research.48

Percutaneous sacroplasty is established to be a safe procedure and provides good short-term and long-term outcomes for patients with painful insufficiency sacral fractures or pathological sacral lesions.28,49 A multicentre study of 243 patients reported mean VAS scores improving from 9.2 to 1.9 for sacral insufficiency fractures and 9.0 to 2.6 for sacral lesions, at 1-year follow-up interval posttreatment.28 However, patients with displaced or unstable sacral fragility fractures would benefit more from surgical fixation.50

In extraspinal bone metastases, cementoplasty is commonly performed to reinforce the acetabular roof and restore load transmission across the acetabulum. There is growing evidence demonstrating percutaneous cementoplasty as a safe and effective choice for patients with painful osteolytic pelvic bone metastases, significantly reducing pain and disability while preserving gait function.51–53 Most recently, Park et al54 evaluated percutaneous cement injection in 178 patients with pelvic bone lesions, which achieved significant pain reduction with mean numerical pain scores decreasing from 6.1 (pretreatment) to 2.1 (1 month posttreatment), while maintaining gait function in 68% of recruited patients.

The role of cementoplasty for metastases in long weight-bearing bones is still debated, due to the presence of complex loading forces (tensile and torsion) in addition to compression. Initial studies reported a 1-year pathological fracture rate of 40.6% after cementoplasty of proximal femoral metastases.55 A systematic review by Cazzato et al53 reported that standalone cementoplasty in the long bones is effective to achieve pain alleviation and to improve mechanical function, but fracture is the most frequent complication.

More recently, a systematic review compared the efficacy between cementoplasty alone versus cementoplasty augmented with fixation devices for impending pathologic proximal femoral fractures. Both appeared effective for pain relief and had a similar postintervention fracture rate (5% for cementoplasty alone versus 7% for augmented cementoplasty), with no statistically significant differences.56 Given current evidence, cementation of the long bones can be considered for palliation of immobile patients who are unsuitable surgical candidates.

Effectiveness

There is mounting evidence of cryoablation for bone metastases both for palliation purposes (Figure 7), as well as curative aim in oligometastatic disease,97,111–113 which is defined as 1-5 metastases, where all metastatic sites are considered safely treatable.114 Most recently, the MOTION trial reported rapid and durable pain relief for patients with metastatic bone disease, as well as improved quality of life and maintained functional status over 6 months.112 Percutaneous cryoablation as an alternative or adjunct therapy for selected patients with plasmacytomas has also been reported.115

CT-guided percutaneous cryoablation of a T10 vertebral metastatic lesion. (A) MRI T1W sagittal image revealed a hypointense lesion involving the T10 vertebral body and posterior elements (arrow). It demonstrated hyperintense signal on (B) sagittal STIR image (arrow in B) with avid enhancement on (C) axial postcontrast T1W with fat saturation (arrow in C), encroaching into the spinal canal (arrowhead in C). (D) Under CT guidance, 2 IceRod cryoablation probes (Boston Scientific, USA) were placed within the T10 vertebral lesion (arrow in D) via a posterior approach, with the more cranial probe inserted coaxially through a 13G Arrow needle (arrowheads in D). (E) Two freeze-active thaw cycles were performed, with the first freeze cycle of

3.5-min duration and the second freeze cycle of 2-min duration, with iceball visualization on CT soft tissue window (arrow in E).

Soft tissue sarcomas are rare tumours with multiple subtypes, the most common of which are liposarcomas and leiomyosarcomas.116 There is emerging evidence of cryoablation as a feasible and safe treatment option for these tumours, especially in recurrent or metastatic disease.117,118 Other tumour subtypes treated by cryoablation include abdominal wall endometriosis,119 neurofibromas,120 and gastrointestinal stromal tumours.121

This is especially so for extra-abdominal desmoid fibromatosis (Figure 8).111 Previously, treatment consists of a combination of surgery and radiotherapy.122 The current first-line treatment for symptomatic desmoid tumours is based on medical therapies, including tyrosine kinase inhibitors and chemotherapy (including methotrexate),123 with respective systemic side effects.

Percutaneous cryoablation for desmoid fibromatosis in a 32-year-old patient. (A) MRI axial postcontrast T1W with fat saturation image revealed a heterogeneously enhancing infiltrative predominantly intramuscular mass in the left glutaeal region (arrow), representing desmoid fibromatosis. Under CT guidance, multiple cryoablation probes were deployed to customize a large ablation zone as visualized by ice formation (arrowheads). Posttreatment MRI axial postcontrast T1W with fat saturation image performed at 1 month (C) and 6 months (D) showed reduced enhancing tissue at ablation site with residual enhancement (arrow in D). The patient proceeded to undergo further cryoablation sessions at residual and recurrent sites.

Recent studies have demonstrated cryotherapy as a safe and effective treatment modality for extra-abdominal desmoid tumours, with efficacy similar to those managed via conventional approaches in the short to medium term.124,125

Percutaneous image-guided neurolysis (Figures 9-12) is a safe, efficient, and relatively cost-effective means of managing refractory pain, which may arise from direct tumour invasion or iatrogenic to surgical and radiation therapy.101 Unlike heat-mediated ablation, cryoablation does not disrupt the acellular epineurium or perineurium, reducing risk of neuroma formation and potentially allowing nerve regeneration.126 Cryoneurolysis is also not associated with systemic toxicity occasionally encountered with chemical nerve ablation.127

Cryoablation for Morton metatarsalgia. (A) MRI long-axis T2W fat-suppressed image of the forefoot and (B) MRI short-axis T1W sagittal image at the level of the metatarsal heads revealed a T2w-hypointense (arrowheads in A) and T1w hypointense lesions in the second and third web spaces (arrows in B), in keeping with Morton neuromas. The third web space lesion was treated with cryoablation. (C) Preprocedural CT revealed soft tissue density (arrow) corresponding with MRI images. (D) An IceSeed cryoablation probe (Boston Scientific, USA) was advanced into the lesion (arrow) with adjacent iceball formation (arrowheads).

Conclusion

There have been substantial advancements and mounting evidence of safety and efficacy in percutaneous image-guided procedures for the musculoskeletal system, including cementoplasty and cryoablation techniques. These enable the interventional radiologist to offer a diverse repertoire of options in a multidisciplinary approach to patient management, especially cancer patients with bony metastases.